

2(a)(i)	1 continuous ;	1
2(a)(ii)	any three from: 1 several / many, genes ; 2 additive effect / gene(s) (products) interact ; 3 environment has, big / significant, effect ; 4 (named) environment(al factors) affects gene expression ; 5 e.g. age / diet / disease / parasites / chemicals / mutagens / alcohol in utero, can affect, development / growth / IPD / phenotype ;	3
2(b)(i)	1 <u>regulator</u>(y) (gene) ; plus any two from: 2 gene codes for / makes, transcription factor / activator / inducer or protein is, transcription factor / activator / inducer ; 3 TF / (PAX3) protein / gene product / inducer / activator, binds to DNA ; 4 (binding site for TF is) promoter / enhancer ; 5 helps / stimulates, <u>RNA polymerase</u> (binding / action) ;	3
2(b)(ii)	any five from: 1 get (total) mRNA from (embryonic) cell (expressing PAX3) ; 2 make, cDNA / ssDNA from mRNA ; 3 add fluorescent, label / tag / dye / chemical, to (this) DNA ; 4 microarray / chip, has DNA, spots / probes / reporters / oligos, from, known / named, genes ; 5 cDNA / ssDNA, hybridises with / binds to / complementary base pairs with, (spots / probes / reporters / DNA, on) chip ; 6 fluorescent spots, indicate / show, (gene / DNA) expression ; 7 AVP ; e.g. ref. to wash off excess (c)DNA after hybridisation UV / laser (scan), to see fluorescence intensity of fluorescence gives quantitative measure	5
2(b)(iii)	any three from: 1 small genetic difference / one gene / PAX3, gives, many / big, phenotypic effect(s) ; 2 because PAX3, regulates / controls, many / other, genes ; 3 different, level / magnitude / intensity / pattern, of gene expression (of other genes in chimps) ; 4 higher PAX3 (expression / protein) makes, chimps eyes closer together / chimp IPD smaller / decreases face width ;	3